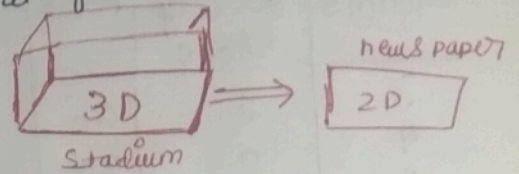


→ In PCA, Original feature → features } called Principal Components

⇒ These components :-

- Linear Combination of original features
- mutually uncorrelated
- Ordered by importance



Example / Analogy :- A photographer capturing a photo in a cricket stadium, which is 3D. The next day he want to print those photos in the news paper, which is 2D, when the people sees these picture they will get same information as from the original stadium view

What are Principal Components :-

→ Principal Components (PCs) are :

- New axes obtained by rotating the original coordinate system and the directions where the data varies most.

Properties :-

- PC1 ⇒ Direction of maximum variance
- PC2 ⇒ Next highest variance, orthogonal to PC1
- PCs ⇒ PCs are uncorrelated and orthogonal

Why PCA used / benefits

PCA is used to solve multiple problems caused by high-D data

- i) To reduce dimensionality
- ii) To remove multicollinearity
- iii) To handle curse of dimensionality
- iv) To improve model performance

iii) data visualization :- Ex. If data has some 20 features.

Visualize these data in 2D, 3D is not possible / interpretable
• because we human can see in 3D, more dimension is not possible

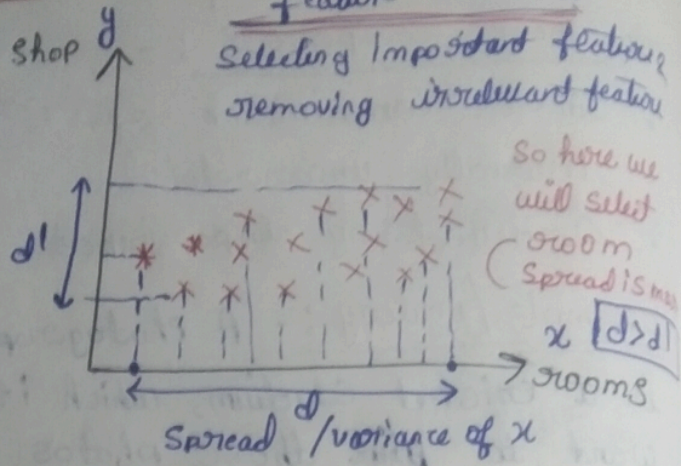
when to use PCA :-

- Many features
- Feature are correlated
- Distance based model
- Visualization required
- preprocessing step.

Geometric Intuition :-

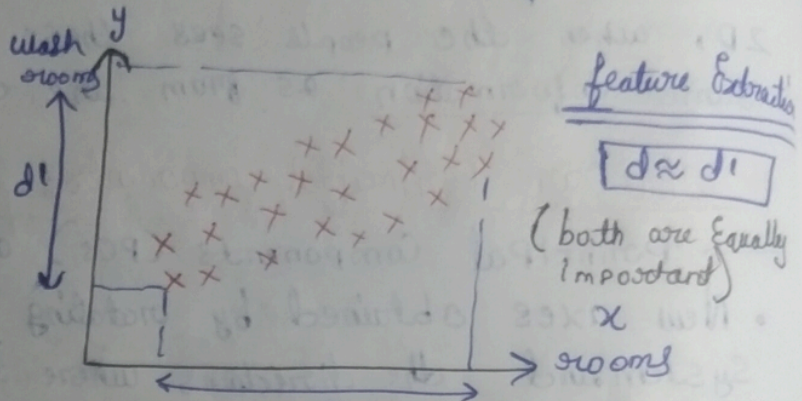
Example 1 :-

rooms	grocery Shops	Price
3	2	60
4	0	90
5	6	170
2	10	125

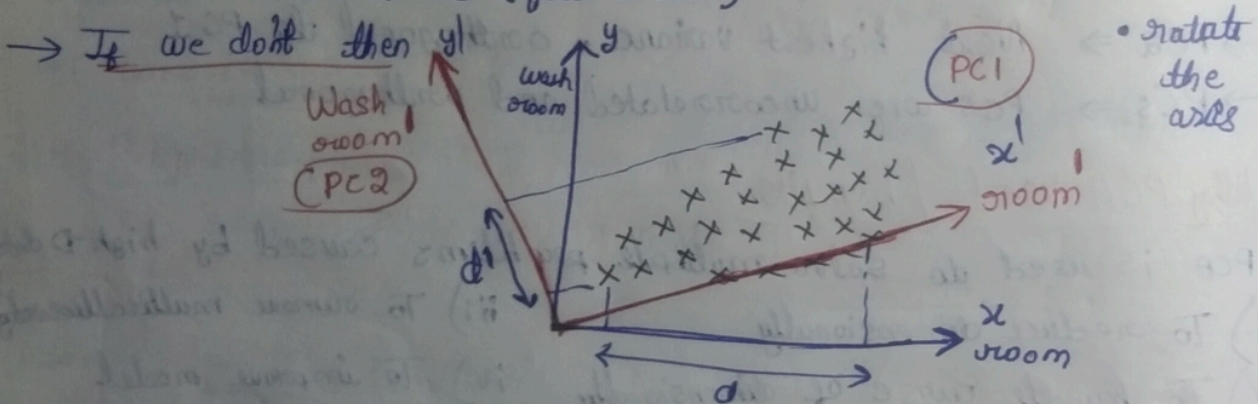


Example 2 :-

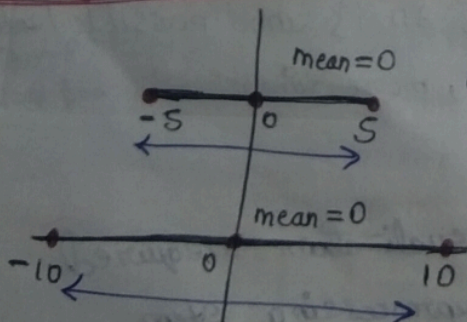
No. of room	Wash rooms	Price
3	4	86
4	3	75
2	2	50
5	4	120



→ If we have domain knowledge, we will combine both columns to one column named (flat size)



Why Variance is important :-



mean may be same, but spread of data is different

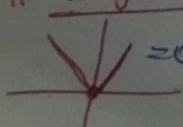
$$\text{Variance} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

→ Variance is proportional to Spread but not Exactly Spread

$$\Rightarrow V_1 \Rightarrow \frac{25 + 0 + 25}{3} = \frac{50}{3}$$

$$V_2 \Rightarrow \frac{100 + 0 + 100}{3} = \frac{200}{3}$$

why not mod :- (mean absolute deviation)

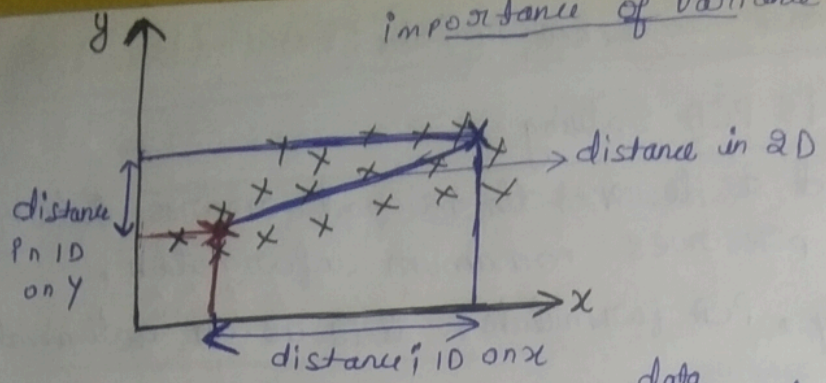


$\Rightarrow$ mod is not differentiable at some point / 0.

$\Rightarrow$ when applying optimization algorithms it's not possible.

tion:-
 at features
 part feature
 So here we
 will select
 room
 spread is more
 x $|d > d|$
 rooms

Importance of Variance

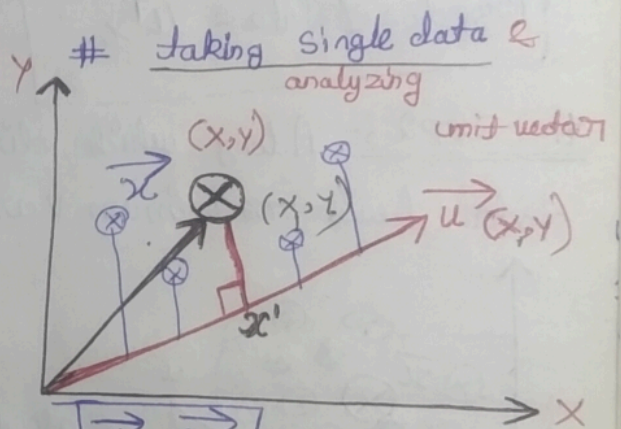
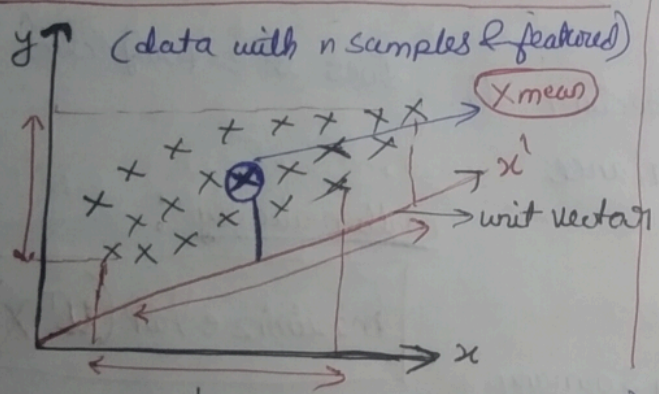


- Variance measures how much spread out data
- High Variance:- Value changes a lot → Rich information
- Low Variance:- Value changes very little → less information
- * In data:- change → information

PCA assumes that directions with high variance carry more useful information about data structure.

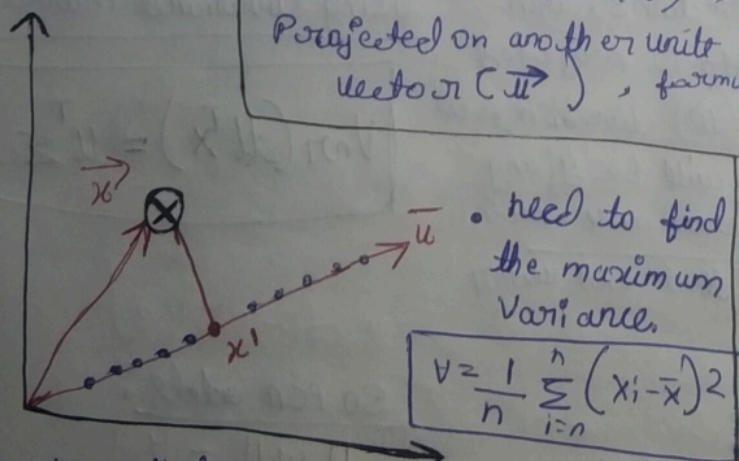
- That's why PCA focuses on variance.
- Variance makes distance meaningful.

Problem Formulation:-



when unit vector ($\vec{x}$) → Projected on another unit vector ($\vec{u}$), formula

$$\frac{\vec{u} \cdot \vec{x}}{|\vec{u}|} = \vec{u} \cdot \vec{x} = \vec{u}^T \vec{x}$$



$$\begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \end{bmatrix} \begin{bmatrix} x_1 & x_2 \\ y_1 & y_2 \end{bmatrix} \Rightarrow x_1 x_2 + y_1 y_2$$

→ when all data points are Projected on unit vector ($\vec{u}$)

$$\sum_{i=1}^n \frac{(\vec{u}^T \vec{x}_i - \vec{u}^T \bar{\vec{x}})^2}{n} \Rightarrow \text{variance}$$

(number) which is scalar (distance)

what problem is PCA Solving:-

- It will find a lower-dimensional representation of data that preserves maximum information.
- Mathematically, PCA formulates this as an optimization problem.

Given:- a dataset with n samples & d features

- data is mean-centered.
- we want to reduce dimensions from $d \rightarrow k$ bidding to small dimension

what PCA wants to find:- PCA wants to find

- A direction (vector) on which, data can be projected.
- Such that the variance of the projected data is maximum.

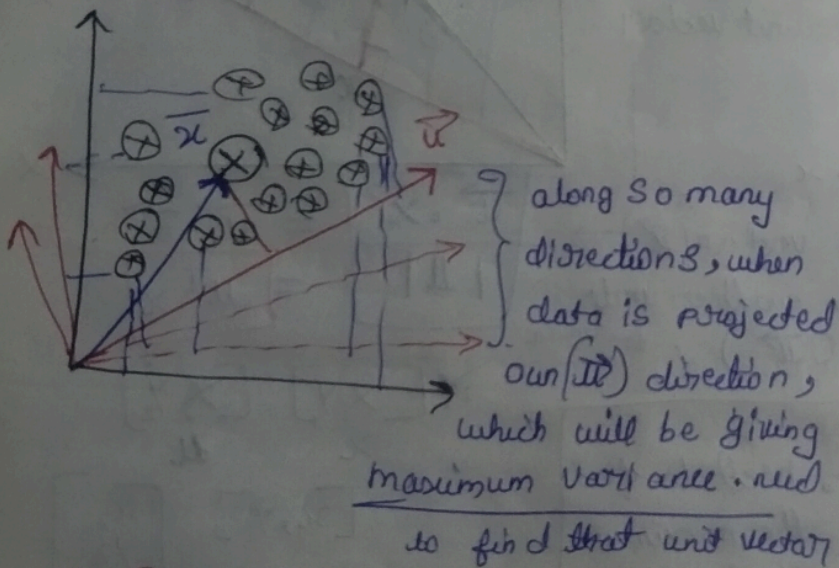
Step 1:- Projection step (understand from previous plots)

- x = original data point
- $u \Rightarrow$ direction (unit vector)

$$\text{Projection of data} = u^T x$$

This converts high-dimensional data into a 1D representation along direction u

Step 2:- Along which direction u , does the projected data have maximum variance



mathematically:

$$\text{maximize } \text{Var}(u^T x)$$

using Covariance Matrix Σ

$$\text{Var}(u^T x) = u^T \Sigma u$$

Constraints:-

without constraints:

- we could scale u infinitely
- Variance become infinite

So PCA adds:

$$\|u\| = 1$$

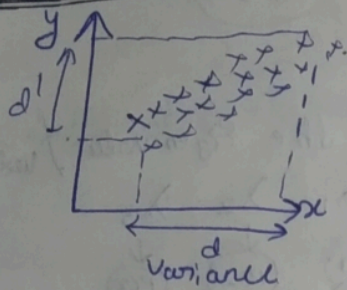
• This Ensures a fair comparison of directions

PCA Optimization Problem:-

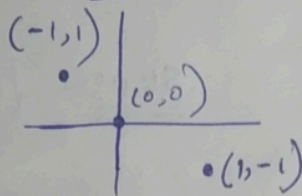
$$\text{maximize } W^T \Sigma W \quad \text{subject to } \|W\| = 1$$

this is the PCA problem formulation

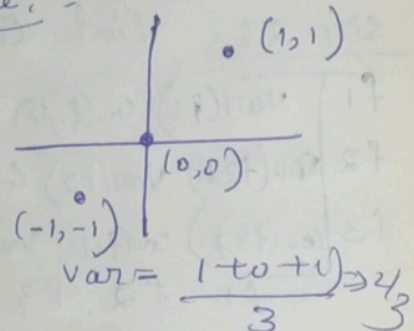
Variance and Covariance:-



Problem of variance:-



$$\text{Var} = \frac{1+0+1}{3} \Rightarrow 2/3$$



$$\text{Var} = \frac{1+0+1}{3} \Rightarrow 2/3$$

→ here for different data, variance is same,
→ from this we don't get to know how variables change together. To solve this Covariance is used

Covariance: - Covariance measures how variables change together.

→ It tells whether two features increase/decrease together or move in opposite direction.

$$\text{Cov}(X, Y) \Rightarrow \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

correlation $\Rightarrow -1$ to 1

+ve $\Rightarrow X \uparrow, Y \uparrow$ together

-ve $\Rightarrow X \uparrow, Y \downarrow$

zero $\Rightarrow$ zero No linear relationship.

Covariance matrix:-

$$\begin{matrix} x_1 & | & x_2 \end{matrix} \Rightarrow \begin{matrix} \uparrow x_2 \\ \rightarrow x_1 \end{matrix} \Rightarrow \begin{matrix} x_1 & x_2 \\ \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) \end{bmatrix} \end{matrix}$$

$\Rightarrow$ Symmetric

$\Rightarrow$ Square matrix

$$\begin{matrix} \text{Cov}(a, b) \\ = \text{Cov}(b, a) \\ \text{Symmetric} \end{matrix} \Rightarrow \begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_2, x_1) \\ \text{Cov}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix} \begin{matrix} x_1 & x_2 \end{matrix}$$

$$\begin{cases} \text{Cov}(x_1, x_1) \\ \downarrow \\ \text{Var}(x_1) \end{cases}$$

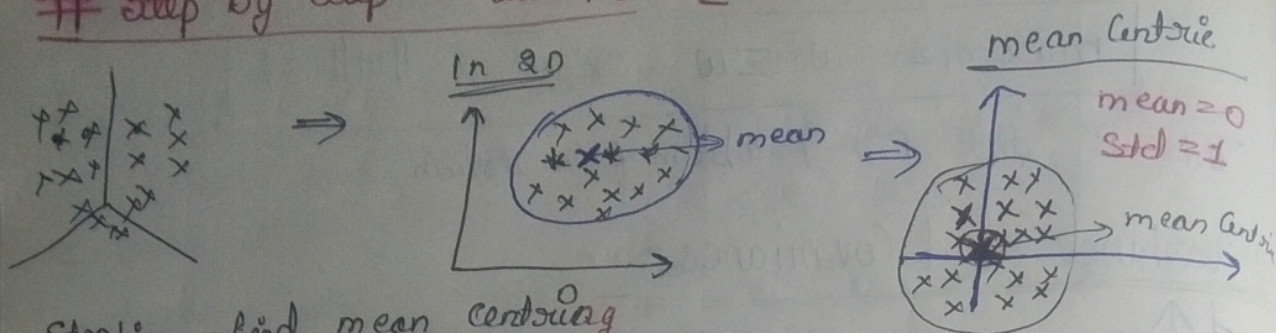
→ In Covariance Matrix, Diagonal Elements are Variance

(variance) → Non-diagonal elements $\Rightarrow$ Symmetric

• Covariance matrix captures Spread of Each feature.

• Relationship b/w features (Covariance).

Step by Step Solution :- [How PCA Solves Problem]



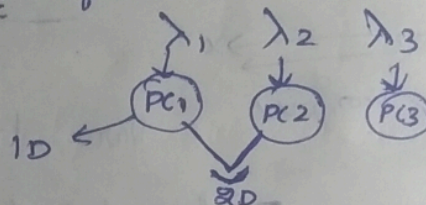
Step 1:- find mean centering

Step 2:- find Covariance matrix

$$\begin{matrix} f1 \\ f2 \\ f3 \end{matrix} \begin{bmatrix} \text{var}(f1) & \text{cov}(f1, f2) & \text{cov}(f1, f3) \\ \text{cov}(f2, f1) & \text{var}(f2) & \text{cov}(f2, f3) \\ \text{cov}(f3, f1) & \text{cov}(f3, f2) & \text{var}(f3) \end{bmatrix}$$

$f1 \quad f2 \quad f3$

Step 3:- find the Eigen value/vector



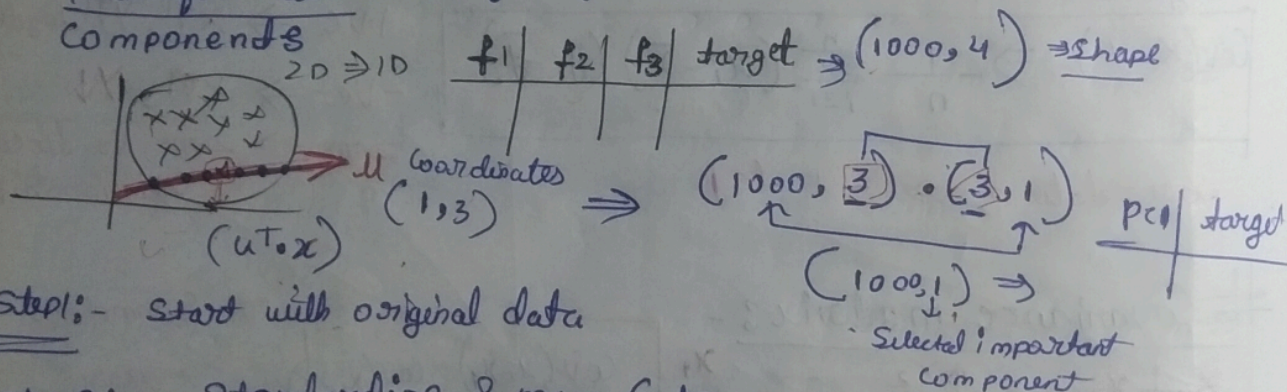
Step 4:- Sort Components by importance

Step 5:- Select top K components

Step 6:- project data on new space (from 3D to 2D)

How PCA Transforms Data Points :-

Transformation = projecting original points onto principal components



Step 1:- start with original data

Step 2:- Standardize & mean Center

$$X_{\text{centered}} = X - \mu$$

→ PCA works on Centered data

Step 3:- Component Principal Components

from Covariance matrix, calculate Eigenvectors and then select top K eigenvectors,

Step 4:- Project points (This is the transformation)

$$Z = \mu^T \cdot X_{\text{centered}}$$

Z = transformed data

from (2D) to 1D ⇒ (100, 3) ⇒ (100, 1)

for one data point x :

$$z = u^T \cdot x$$

- Dropping a $\perp$ from ~~to~~ point to PC axis.
- Measuring its coordinate along that axis
- Think of shadow of point on new axis



What is Explained Variance:-

Explained variance tells us, How much information (spread/variability) of the original data is captured by each principal component.

Variance = Information

Ex:- Imagine MNIST digits:

- If a pixel never changes → useless
- If a pixel changes a lot across images → Informative
- Variance measures how much things change.

So PCA: Keeps directions where data varies the most
drops directions where data is almost constant.

After PCA, for each principal Component:

$$\text{Explained Variance} = \lambda$$

$\lambda \Rightarrow$ Eigen value of Covariance matrix

Bigger Eigen value → more variance captured

Explained Variance Ratio:-

Usually we don't care about raw variance, we care about percentage.

$$\text{Explained Variance Ratio} = \frac{\lambda}{\sum_{i=1}^n \lambda_i} \Rightarrow [\lambda_1 + \lambda_2 + \dots + \lambda_n]$$

[Percentage per Component]

Cumulative Variance

np.cumsum(pca.Explained-Variance-Ratio-)

[0.25, 0.15, 0.10, 0.07, ...]

Cumulative $\Rightarrow$ [0.25, 0.40, 0.50, 0.57, ...]

• first 2 PCs $\Rightarrow$ 40% info

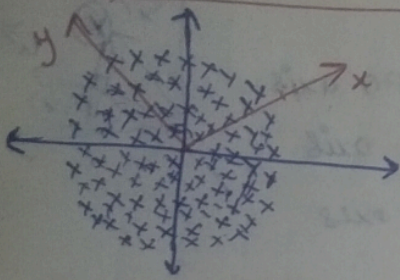
• first 10 PCs $\Rightarrow$ 75% info

• first 50 PCs $\Rightarrow$ 90% info

PCA (n-components = 0.95)

Keep min no. of PCA components that Explains 95% Variance

When PCA does not work :-



i) non linear data

- PCA can only draw straight lines
- No straight line can separate a circle properly.

result

- PCA projection overlap points
- structure is lost.

ii) Low Variance but important features :-

Ex: Fraud detection.

→ 99% transactions → normal → 1% → fraud

- So Fraud Features appear rarely, low variance but are crucial

what PCA does :-

- keeps high-variance, normal behaviours
- Removes low-variance fraud signals

result:- accuracy drops badly.

iii) Supervised tasks → PCA ignores labels :-

Two classes

Class A : • • • • •

Class B : • • • • •

→ If both classes spread in the same direction

- PCA keeps that direction but it does not separate classes

result:- High variance, poor classification

(Better LDA)

iv) Unscaled features :- dominated by large-scale features

Ex: age (0-100):

Salary (0-10,00,000)

Salary variance >> Age variance

- PCA focus on salary, age information ignored

vi) Noisy data :- Ex: Sensor data

- Signal variance = low → PCA treats noise as 'important',
- Noise Variance = high → keep noisy components, model overfit

v) Small dataset :- Unstable Covariance

• Samples = 20

• features = 1000

• covariance estimated from very little data
Eigen values become unreliable.

